

Architecture Quality — 9.5 / 10
Enterprise Readiness — 9 / 10
Technical Depth — 10 / 10
Business Strategy — 8.5 / 10 (revised)

SOFTWARE REQUIREMENTS SPECIFICATION

FinFlow Terminal

The AI Operating System for Global Capital
Markets — an autonomous financial
intelligence platform, not a Bloomberg clone.

Version 2.0.0 · 2026-07-03

Prepared by the Architecture & AI Strategy Team

Status: Revised — Incorporates AI-native strategic addendum

Revision Notice

What changed in v2.0.0. An internal architecture review (attached as Appendix B) scored the original SRS 9.5/10 on architecture but only 5.5/10 on business strategy, concluding that the platform was designed as "*Bloomberg + AI*" rather than an AI-native category of its own. This revision keeps essentially all of the original v1.0.0 architecture — the microservice layout, data model, event backbone, and roadmap remain ~70% intact — but reframes the product vision and adds twelve missing capability areas (Section 15) that shift FinFlow from a terminal-with-AI-features into an autonomous financial intelligence platform.

Revised vision statement: "Build the world's first autonomous financial intelligence platform where a portfolio manager can ask a question in natural language, and a network of specialized AI agents gathers market data, internal documents, macro events, company fundamentals, alternative data, and quantitative models to produce a fully explainable investment recommendation."

Change	Description
Positioning	Removed "Bloomberg alternative" framing throughout; repositioned as "The AI Operating System for Global Capital Markets."
De-prioritized	Plugin marketplace, Matrix chat, notebook environment, and Bloomberg-skin command palette are retained but reclassified from flagship differentiators to <i>supporting</i> features (see Section 6.1 priority changes).
Added	Twelve new capability areas — Enterprise Memory, AI Research Agents, Knowledge Graph, Simulation Engine, Explainable AI, Sovereign/On-Prem AI, Agent Marketplace, Autonomous Workflows, Voice Interface, Multi-Modal Ingestion, Financial Digital Twin, Enterprise Decision Engine — detailed in Section 15.
Roadmap	New "AI Intelligence Layer" and "Autonomous Platform" phases inserted into the 18-month roadmap (Section 10).

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1. Introduction

FinFlow Terminal delivers an **AI-native financial intelligence platform** for global capital markets. Rather than replicating the Bloomberg Terminal's feature checklist, FinFlow is built around a network of specialized AI research agents operating over real-time market data, enterprise documents, and a financial knowledge graph — producing explainable, evidence-backed investment recommendations on demand.

This SRS captures **what** the system must do, **how** it must behave under various constraints, and the architectural blueprints needed for implementation, testing, and future evolution. Section 15 is a strategic addendum documenting capability areas identified in the v1.0.0 architecture review as necessary to differentiate FinFlow from prior Bloomberg-alternative attempts (Refinitiv, FactSet, Koyfin, Symphony, AlphaSense, Sentieo).

2. Scope

In-Scope	Out-of-Scope
Cross-platform client (Web SPA, Desktop, iOS/Android) · Real-time market data ingestion & distribution · News & social-sentiment feeds · AI research agent network & enterprise memory (Section 15) · Secure federated messaging · Trading adapters (FIX, REST, WebSocket) · Risk engine (VaR, CVaR, stress testing) · Reference-data service (FIGI, LEI, ISIN) · Knowledge graph & simulation engine · Agent marketplace · Tiered SaaS + on-prem/air-gapped option	Direct exchange membership or clearing-house integration (outsourced to partners) · Proprietary hardware · Full legacy Bloomberg chat bridge (best-effort optional bridge only) · Physical data-center procurement

3. Definitions, Acronyms & Abbreviations

Acronym	Meaning
API	Application Programming Interface
DFD	Data Flow Diagram
ER	Entity-Relationship
EMS	Execution Management System
E2EE	End-to-End Encryption
FIX	Financial Information eXchange protocol
GDPR	General Data Protection Regulation
KG	Knowledge Graph NEW
OIDC	OpenID Connect
RAG	Retrieval-Augmented Generation NEW
RBAC	Role-Based Access Control
SLA	Service Level Agreement
VaR	Value-at-Risk
XAI	Explainable AI NEW

4. References

1. CLAUDE.md — Project coding & versioning guidelines.
2. Bloomberg Terminal feature baseline (industry reference).
3. Matrix.org — Open secure messaging protocol.
4. Apache Kafka / Flink documentation — Stream processing.
5. PostgreSQL & ClickHouse official docs — Storage engines.
6. TensorFlow Serving / TorchServe / vLLM — Model serving frameworks.
7. OAuth 2.0 & OpenID Connect Core — Auth standards.
8. ISO/IEC 27001, SOC 2 Type II — Security & compliance frameworks.

9. MITRE ATT&CK for Finance — Threat model reference.
10. MiFID II, EMIR, Dodd-Frank — Regulatory scope.
11. Internal Architecture Review, "Bloomberg + AI vs. AI-Native Platform," 2026-07-03 (Appendix B).

5. Overall Description

5.1 Product Perspective

FinFlow Terminal is deployed as multi-tenant SaaS or private-cloud/on-prem/air-gapped installation. It consists of loosely coupled microservices communicating via REST, GraphQL, gRPC, and WebSocket, with an event-driven backbone (Kafka), plus an **AI Intelligence Layer** (agent orchestrator, enterprise memory, knowledge graph, model serving) that sits above the core services and reasons across them — see Section 7.1.

5.2 Product Functions

Function	Description
Market Data Service	Ingests, normalizes, and distributes real-time & historical prices, fundamentals, reference data.
News & Sentiment Service	Aggregates licensed news wires and open feeds; runs NLP sentiment models.
Analytics & Notebook Service	Hosts JupyterLite-style notebooks for custom quant code (supporting feature).
Trading / EMS Service	Order ticket UI, FIX adapter, execution simulation, trade persistence.
Risk & Compliance Service	VaR, CVaR, stress testing; position-limit monitoring; audit logs.
Messaging & Collaboration	Matrix-based secure chat (supporting feature, optional Bloomberg-chat bridge).
Reference Data Service	Central FIGI/LEI/ISIN repository with query and change-request workflow.
Agent Marketplace	NEW — replaces Plugin Marketplace as flagship extensibility model (Section 15.7).
AI Intelligence Layer	NEW — Agent orchestrator, enterprise memory, knowledge graph, simulation engine (Section 15).
Admin & Billing Portal	Tenant management, usage metering, subscription plans, SLA monitoring.

5.3 User Classes & Characteristics

User Class	Primary Goals
Trader / Sales-Desk	Low-latency quotes, one-click trading, instant chat, fast answers instead of more charts.
Analyst / Quant	AI research agents, enterprise memory search, knowledge-graph reasoning, back-testing.
Risk Officer	Real-time risk dashboards, limit alerts, explainable model outputs, audit-trail export.
Portfolio Manager	Autonomous morning briefs, scenario simulation, evidence-backed recommendations.
Compliance / Legal	GDPR/MiFID II adherence, consent management, encrypted communications, explainability records.
IT / Ops	Kubernetes deployment, observability, RBAC, air-gapped/on-prem installs.
Data Provider	Secure ingest API, SLA monitoring, usage-based billing.
Agent / Plugin Developer	SDK, sandbox, versioned marketplace, revenue share for published agents.

5.4 Operating Environment

Clients: Web (Chrome/Firefox/Edge/Safari), Desktop (Windows 10/11, macOS 13+, Linux), Mobile (iOS 16+, Android 13+). Server-side: Linux (Ubuntu 22.04 LTS / RHEL 9) on Kubernetes (managed EKS/GKE/AKS, on-prem, or **air-gapped** per Section 15.6). Network: TLS 1.3 external, mTLS internal. Storage: AES-256 at rest, geo-distributed object store.

5.5 Design and Implementation Constraints

- **Modularity:** No service imports code from another provider's specific module (CLAUDE.md Shared Utilities principle).
- **Zero-Defect Philosophy:** Every production change requires unit, integration, contract, and security tests enforced in CI.
- **Versioning:** Any production-file change triggers a semver bump and lockfile update in the same commit.

- **Performance:** 95th-percentile tick-to-display latency ≤ 150 ms (premium tier); agent-answer p95 ≤ 20 s for multi-source queries (Section 15.2).
- **Scalability:** Horizontal scaling to $\geq 100k$ concurrent users.
- **Security:** E2EE chat, data-at-rest encryption, regular penetration testing, SOC 2/ISO 27001 readiness.
- **Compliance:** Audit-log export (JSON/CEF), data-subject request APIs, data-residency controls.

6. Specific Requirements

6.1 Functional Requirements

Must Should Could New in v2.0

ID	Requirement	Priority
FR-001	User shall log in via email/password, SSO (OIDC), or API key (MFA optional).	Must
FR-002	System shall display real-time price ticks ≤ 150 ms (premium) / ≤ 500 ms (standard).	Must
FR-003	User shall create & manage shareable watchlists with alerts.	Must
FR-004	System shall provide a filterable news feed with real-time sentiment scores.	Should
FR-005	System shall embed an interactive notebook (Python/R/Julia) for live data queries.	Could (demoted from Should)
FR-006	System shall support charting with custom indicators via JS/Wasm plugins.	Should
FR-007	System shall provide a trading ticket UI with FIX adapter order submission.	Must (promoted from Could)
FR-008	System shall run risk calculations (VaR, CVaR, stress) and return results ≤ 2 s.	Should
FR-009	System shall provide secure chat (Matrix, E2EE) with optional Bloomberg-chat bridge.	Could (demoted from Should)
FR-010	System shall expose a reference-data service (FIGI, LEI, ISIN) via GraphQL.	Must
FR-011	System shall provide an admin portal for tenant/billing/SLA management.	Must
FR-012	System shall support an Agent Marketplace for signed, sandboxed AI agents, data adapters, and algos (supersedes Plugin Marketplace).	Must (promoted, redefined — Section 15.7)
FR-013	System shall generate tamper-evident audit logs exportable to SIEM.	Must
FR-014	System shall support GDPR/CCPA data-subject requests.	Must
FR-015	System shall provide configurable API rate limiting and usage analytics.	Should
FR-016	System shall support active-active HA deployment across ≥ 2 AZs.	Must
FR-017	System shall offer a "Bloomberg mode" UI skin and command palette for migration ease.	Could (demoted from Should)
FR-018	System shall provide mobile push notifications for alerts, news, and mentions.	Should
FR-019	System shall support bring-your-own-data (BYOD) ingest as a first-class data source.	Could
FR-020	System shall support multi-language UI (English, Mandarin, Spanish, French).	Could
FR-021	System shall answer natural-language research questions by orchestrating specialized AI agents over enterprise memory, market data, and reference data, returning a confidence-scored, source-cited report.	Must NEW
FR-022	System shall index enterprise documents (PDFs, filings, transcripts, emails, spreadsheets) into a searchable Enterprise Financial Memory.	Must NEW

FR-023	System shall maintain a financial knowledge graph linking companies, suppliers, executives, patents, litigation, commodities, and countries.	Should NEW
FR-024	System shall run macro/market scenario simulations propagating shocks across currencies, equities, rates, and supply chains.	Should NEW
FR-025	Every AI-generated answer shall include source documents, a confidence score, an evidence chain, and any contradictory or missing information.	Must NEW
FR-026	System shall support on-prem, air-gapped, and BYO-model (Llama, Qwen, DeepSeek, Claude, Gemini, OpenAI) LLM deployment.	Should NEW
FR-027	System shall run scheduled autonomous workflows (e.g., 05:00 overnight brief) with no human intervention required to produce the report.	Should NEW
FR-028	System shall accept voice queries and respond conversationally, in addition to typed commands.	Could NEW
FR-029	System shall ingest and reason over multi-modal inputs: images, PDFs, spreadsheets, slide decks, and earnings-call audio/video.	Should NEW

6.2 Non-Functional Requirements

Category	Requirement	Metric / Target
Performance	Market-data tick-to-display latency	≤150 ms (95th p) premium; ≤500 ms standard
Performance NEW	Multi-agent research query end-to-end latency	≤20 s (95th p) for a 5-agent query
Throughput	Ingest rate from market feeds	≥200k msgs/sec per Kafka topic
Scalability	Concurrent active users	≥100k with auto-scaling; tested to 250k
Availability	System uptime (SLA)	99.9% monthly
Disaster Recovery	RTO / RPO	RTO ≤30 min; RPO ≤5 min
Security	Data encryption	AES-256 at rest; TLS 1.3 in transit; mTLS internal
Explainability NEW	Answer traceability	100% of agent answers link to source evidence
Auditability	Log retention	≥7 years; immutable (WORM) storage
Compliance	Certifications	SOC 2 Type II, ISO 27001, GDPR, CCPA-ready before enterprise launch
Maintainability	Code coverage	Unit ≥80%; Integration ≥70%
Sovereignty NEW	Data residency for on-prem/air-gapped tenants	Zero external network calls in air-gapped mode
Observability	Metrics, logs, traces	Prometheus + Grafana; Loki; Jaeger tracing <5 ms overhead
Portability	Client OS support	Windows 10/11, macOS 13+, major Linux, iOS 16+, Android 13+

7. System Architecture

7.1 Component Diagram (C4 — Container Level)

The architecture retains the original microservice topology and adds an **AI Intelligence Layer** (orchestrator, enterprise memory, model serving) that sits alongside the API Gateway and reasons across every core service.

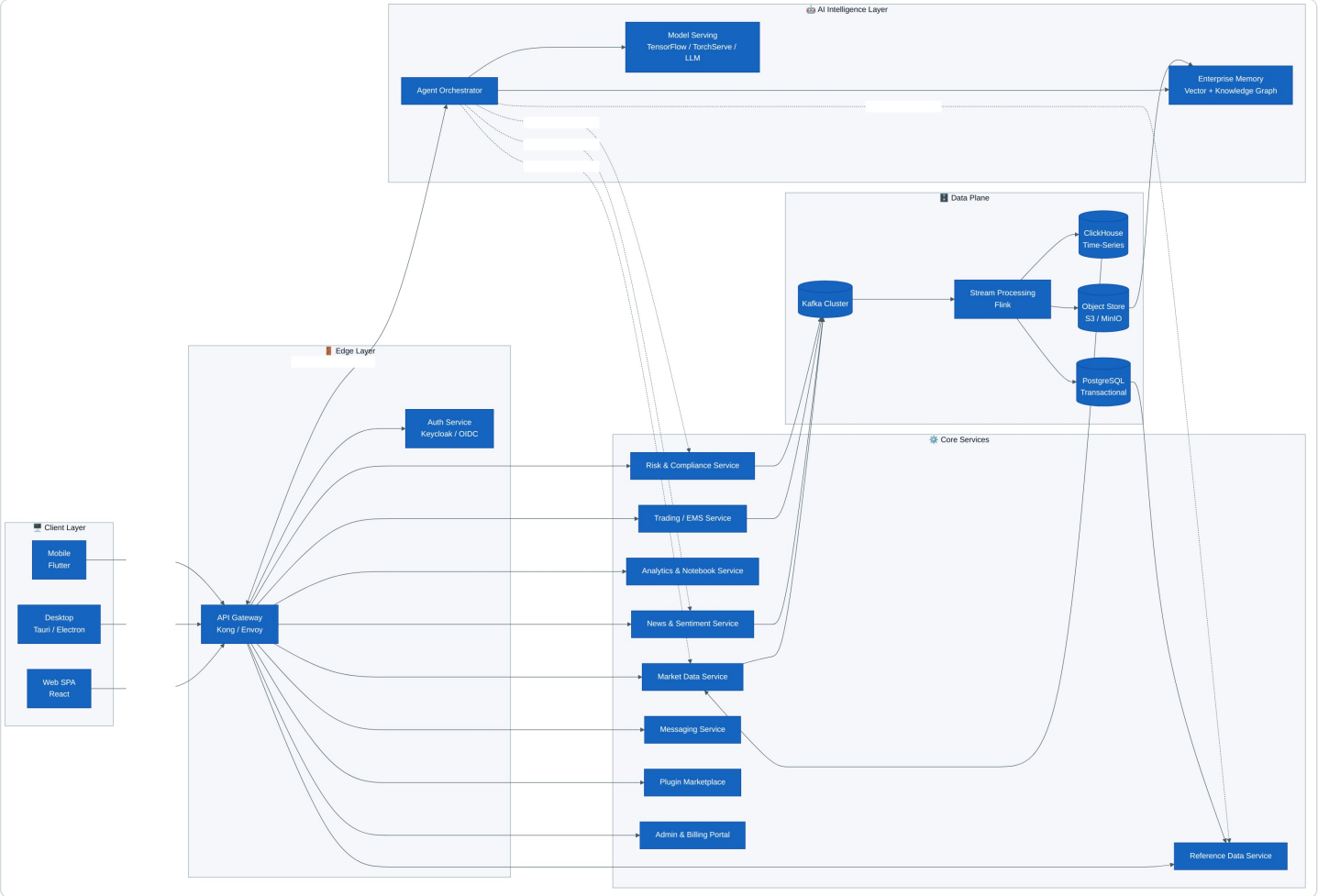


Fig. 7.1 — Container-level component diagram including the new AI Intelligence Layer.

7.2 Data Model (ER Diagram)

New entities — **AGENT_DEFINITION**, **AGENT_RUN**, **EVIDENCE_ITEM**, **KNOWLEDGE_NODE**, and **KNOWLEDGE_EDGE** — extend the transactional schema to persist agent research history, source evidence, and knowledge-graph relationships.

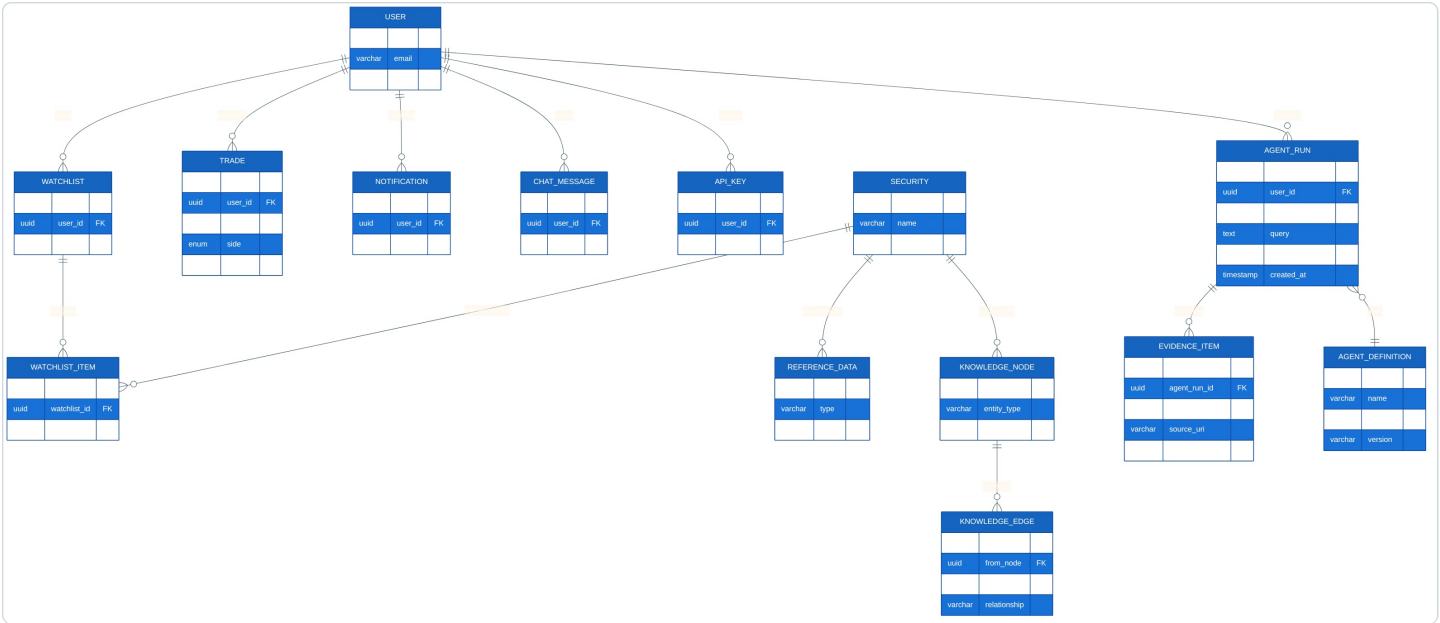


Fig. 7.2 — Entity-relationship diagram with agent and knowledge-graph entities.

7.3 Data Flow Diagrams

7.3.1 Context Diagram (Level 0)

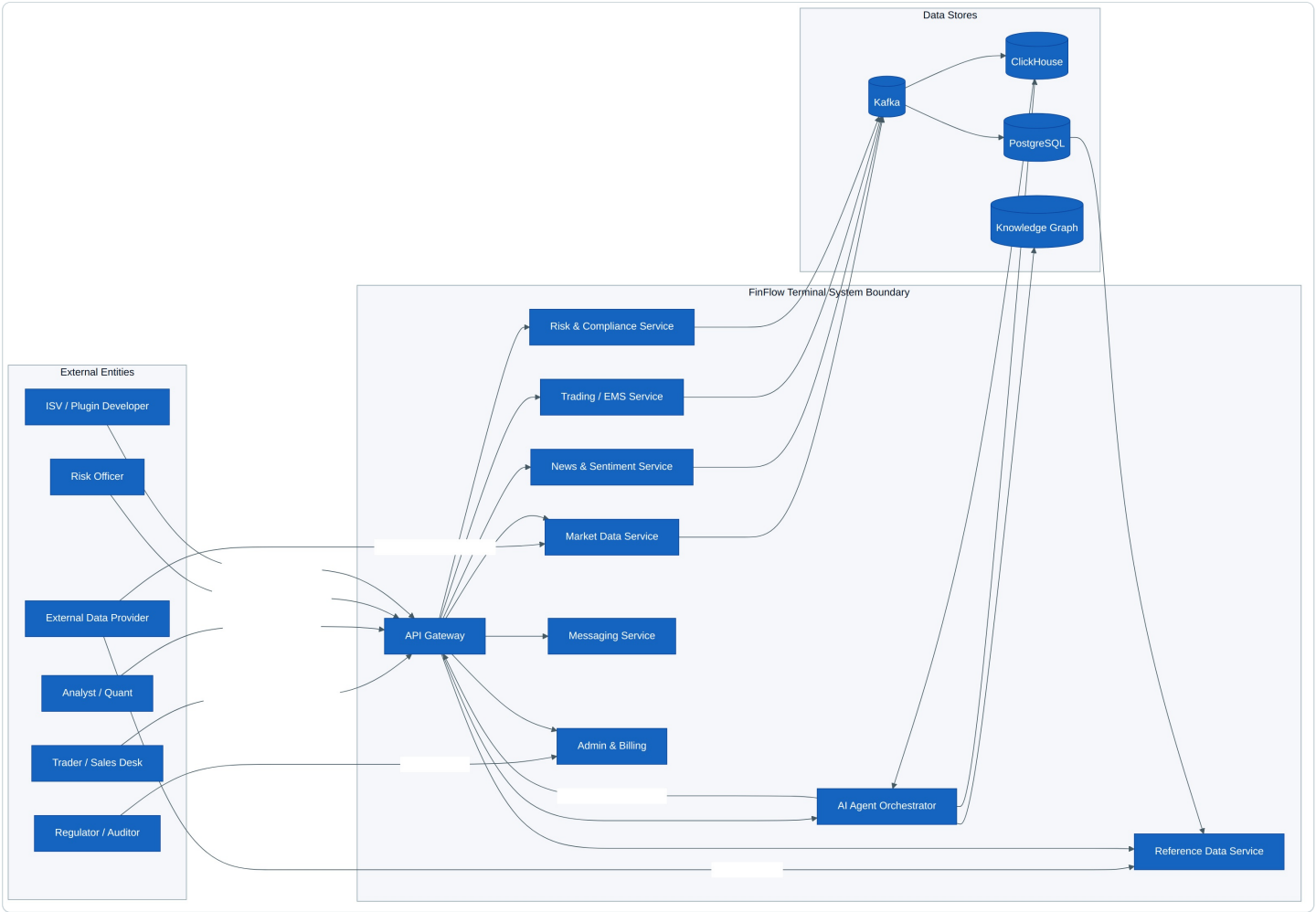


Fig. 7.3 — Context-level DFD showing the Agent Orchestrator as a first-class internal actor.

7.3.2 Level-0 DFD (Core Processes)

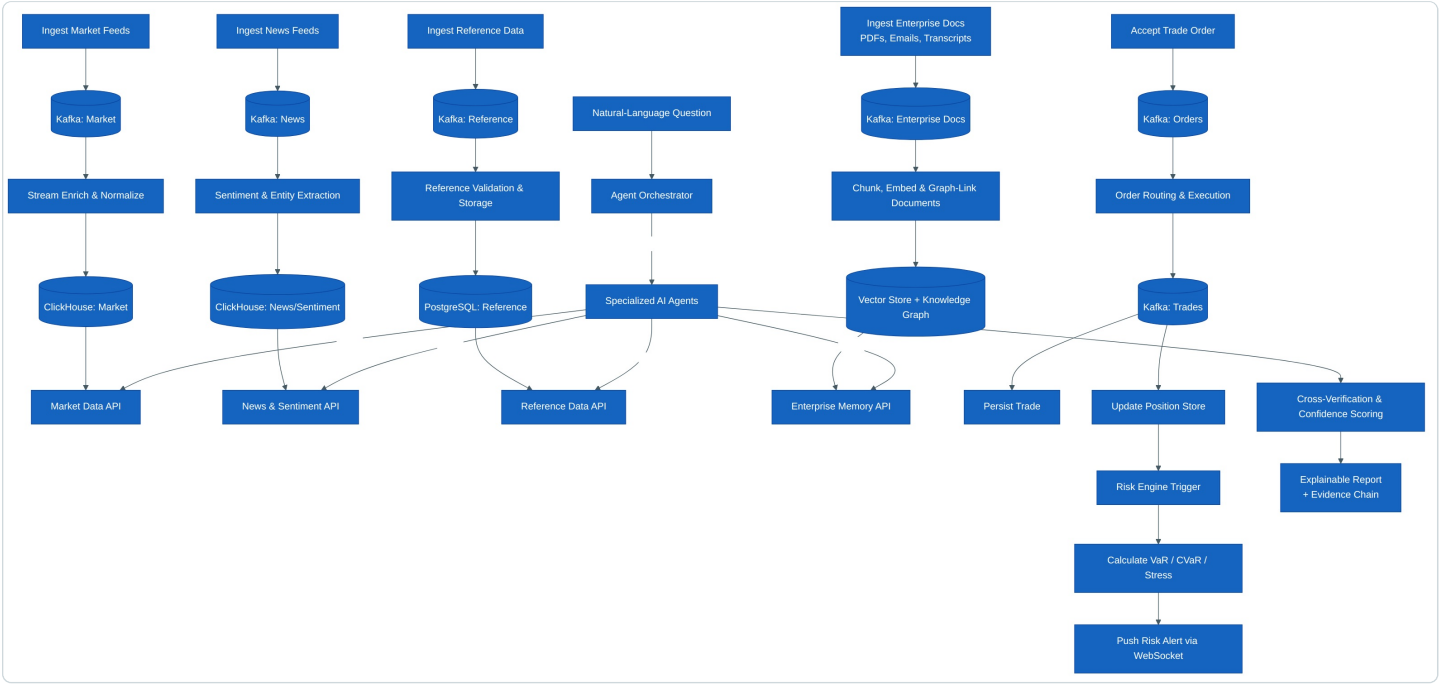
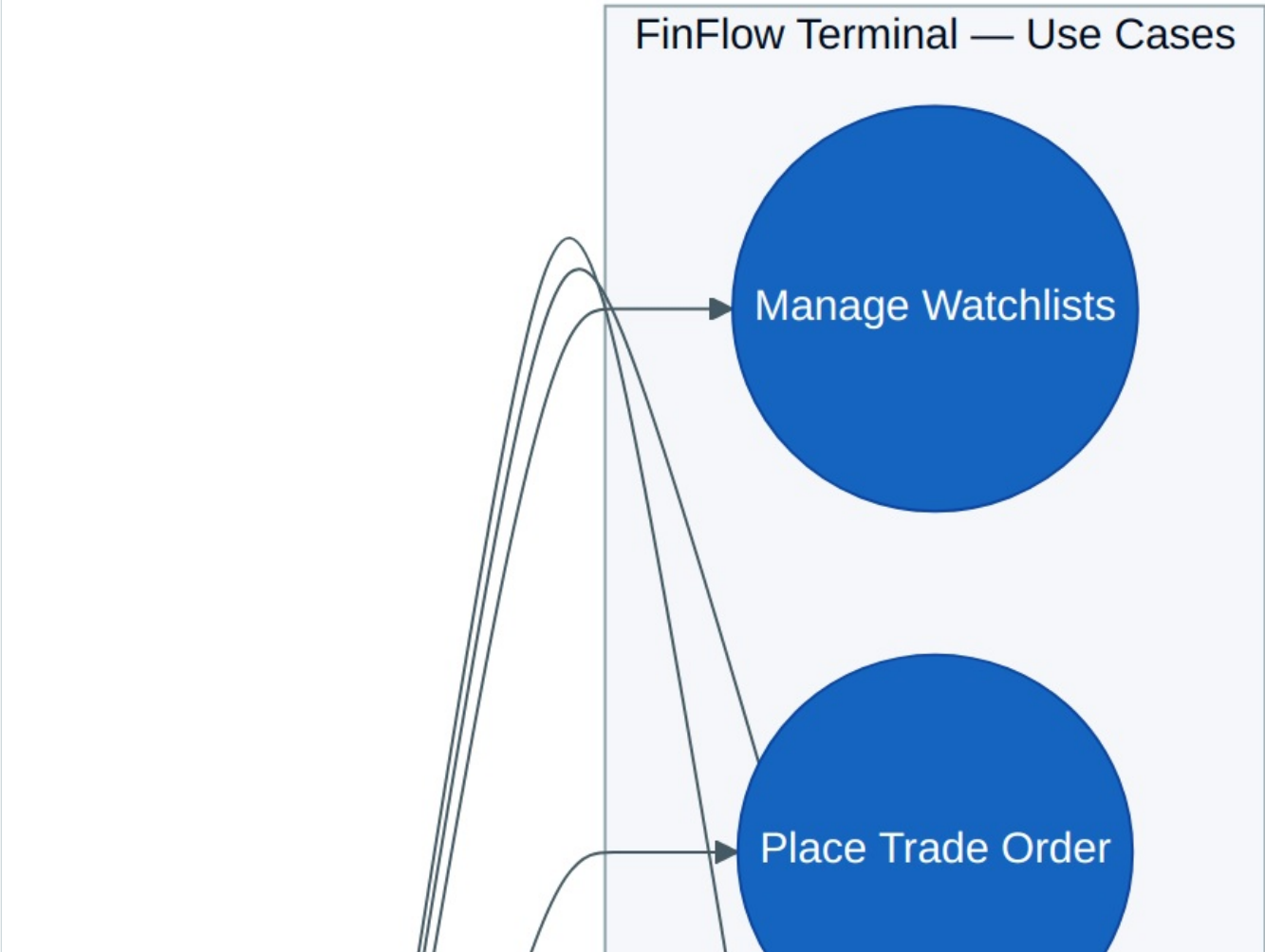
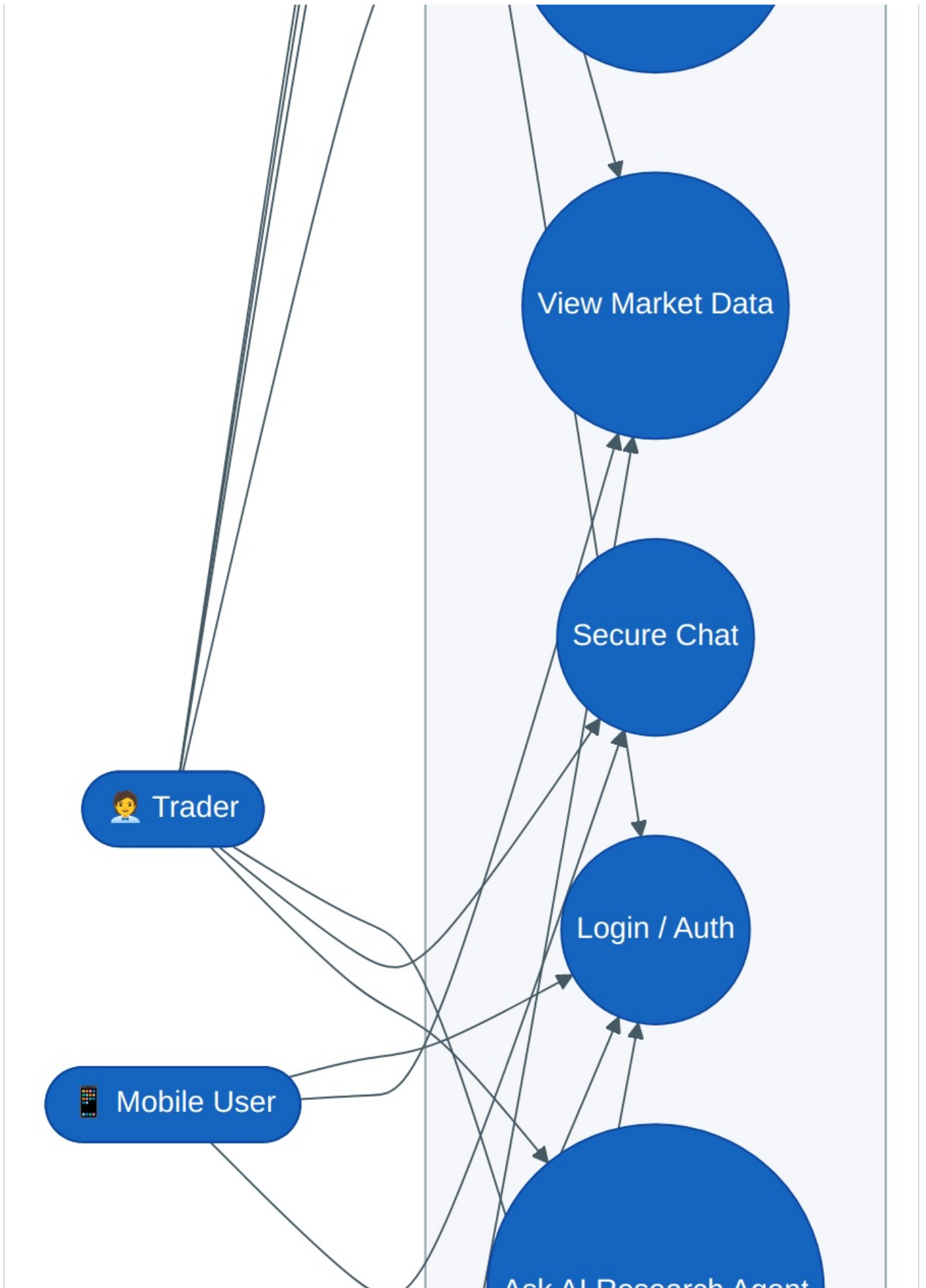
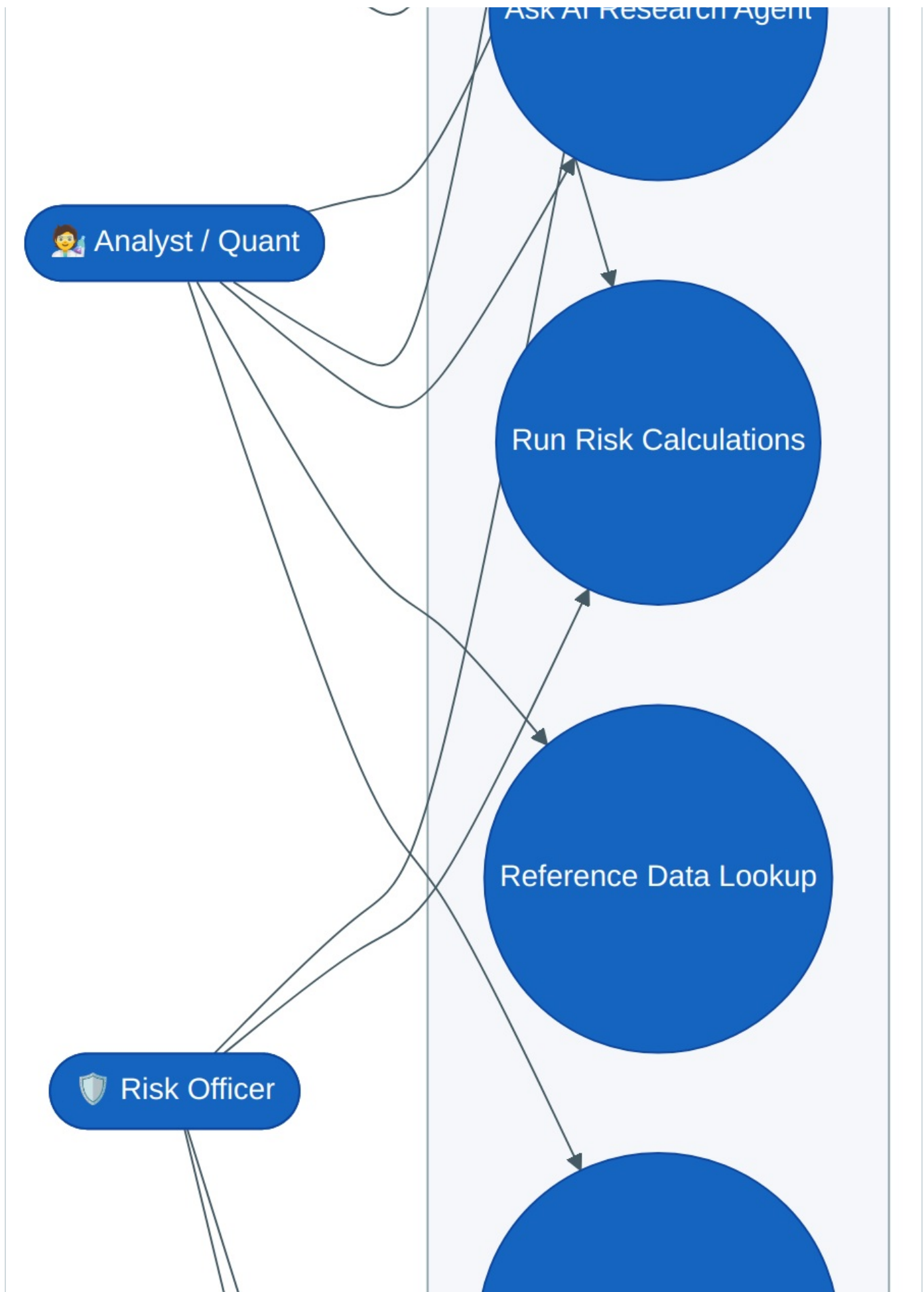


Fig. 7.4 — Core process DFD including enterprise-document ingestion and the agent decomposition pipeline.

7.4 Use-Case Diagram







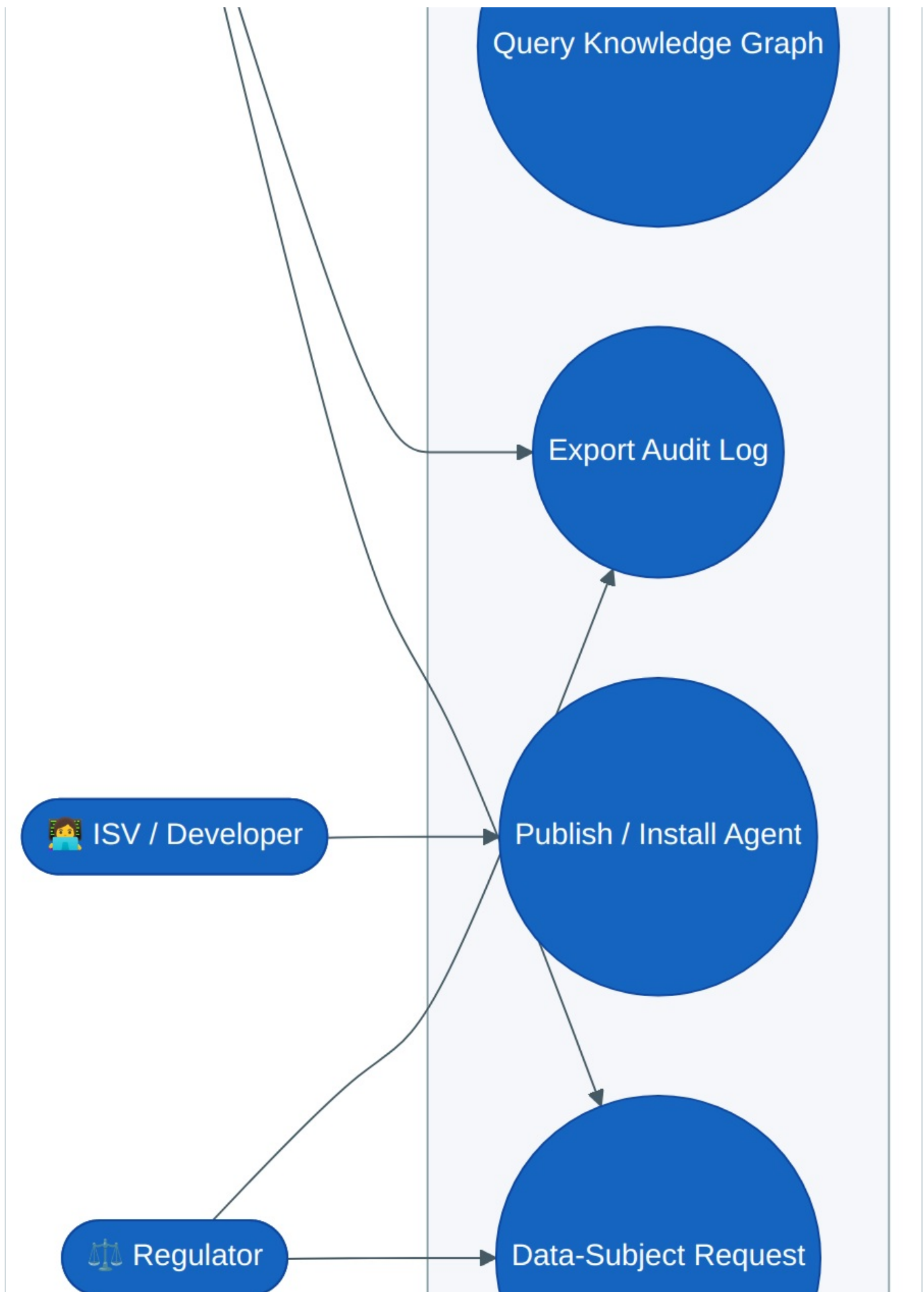




Fig. 7.5 — High-level use-case diagram, including "Ask AI Research Agent" and "Query Knowledge Graph."

7.5 Sample Sequence Diagram — Real-Time Price Tick Flow

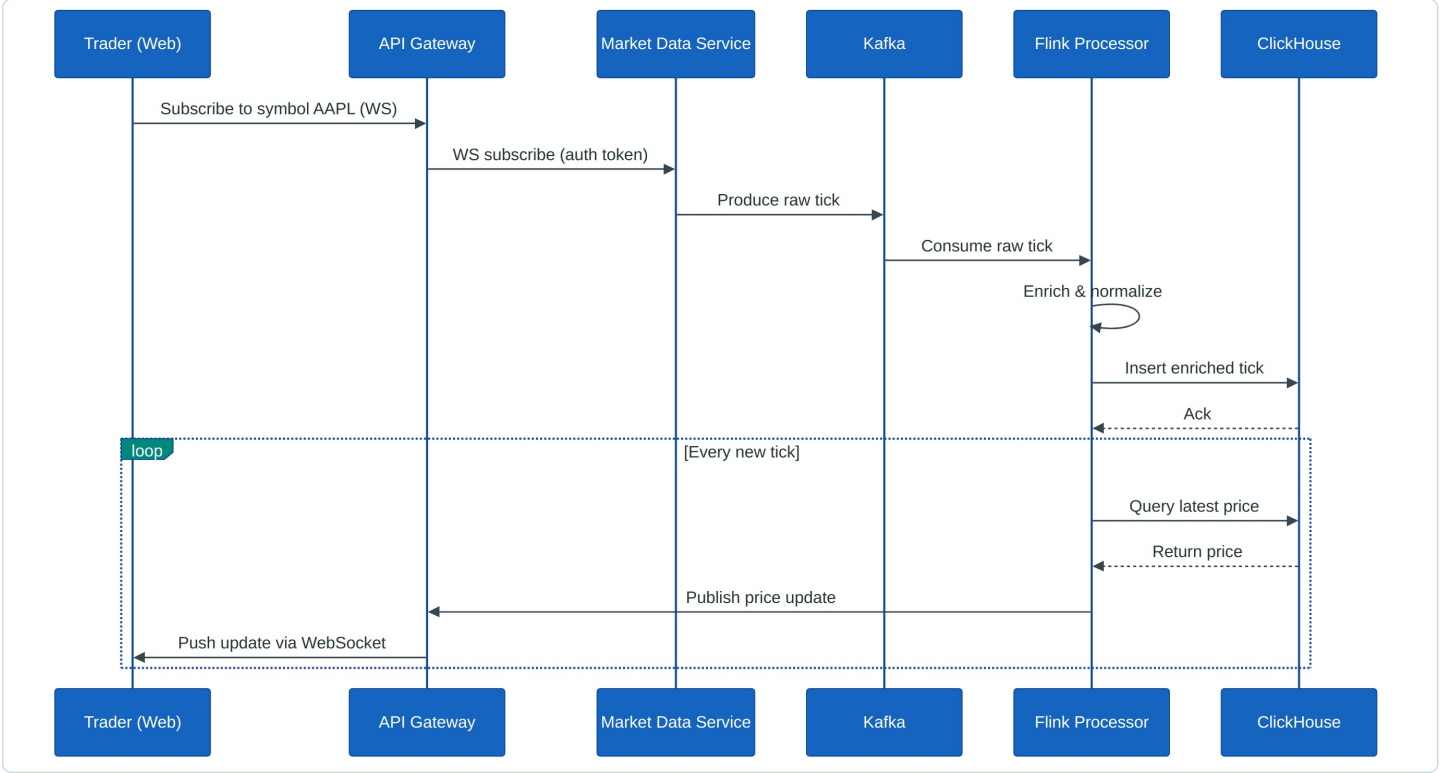


Fig. 7.6 — Tick ingestion, enrichment, and WebSocket push sequence.

8. External Interface Requirements

Interface	Description	Protocols / Standards
User Interface	Web SPA, Desktop, Mobile	HTML5, TypeScript, WebSocket, REST/GraphQL
API Gateway	AuthN, AuthZ, rate-limit, routing	TLS 1.3, HTTP/2, OAuth2/OIDC, API-Key
Market Data Ingress	Exchange/vendor/alt-data feeds	FIX, FAST, WebSocket, HTTPS, MQTT (optional)
Reference Data Ingest	FIGI, LEI, ISIN providers	SFTP, HTTPS, S3, Kafka Connect
Trading/EMS Outbound	Order routing to brokers/exchanges	FIX 5.0/SP2, REST, WebSocket
Messaging	Matrix federation + optional Bloomberg-chat bridge	Matrix client-server API, Olm/Megolm
Agent Marketplace NEW	Upload/download signed agents & datasets, sandbox execution	HTTPS, OAuth2, JWT, WASM sandbox
Enterprise Memory API NEW	Document ingest, semantic search, evidence retrieval	REST/gRPC, vector index protocol
Model Serving NEW	LLM inference (cloud + on-prem/BYO model)	gRPC, OpenAI-compatible REST, air-gapped local inference
Data Export / SIEM	Audit logs, compliance reports	Syslog, CEF, JSON over HTTPS
CI/CD	Build, test, deploy pipelines	GitHub Actions, Docker, Helm, ArgoCD, Kubernetes

9. Other Non-Functional Attributes

Attribute	Requirement
Portability	Clients run on listed OS/desktop/mobile platforms; server-side on any CNCF-conformant Kubernetes.
Interoperability	All services expose OpenAPI v3 or gRPC proto; one major version of backward compatibility guaranteed.
Maintainability	Monorepo with clear service boundaries; each service has its own Dockerfile, Helm chart, unit tests.
Testability	100% of public API endpoints have contract tests (Pact); nightly performance benchmarks.
Security	Defense-in-depth: network segmentation, zero-trust mTLS, CSP headers, pen-testing, bug-bounty.
Compliance	Audit-log includes user, timestamp, action, resource, outcome; WORM storage; GDPR data-subject endpoints.
Usability	WCAG 2.1 AA; keyboard-navigable command palette; contextual help.
Extensibility	Agent/Plugin SDK with TypeScript defs, WASM sandbox, semver contract, security scanning before publish.

10. Roadmap & Milestones

The 18-month roadmap below reflects the v2.0.0 plan, including the new AI Intelligence Layer and Autonomous Platform phases. See Section 15.14 for phase-level rationale.

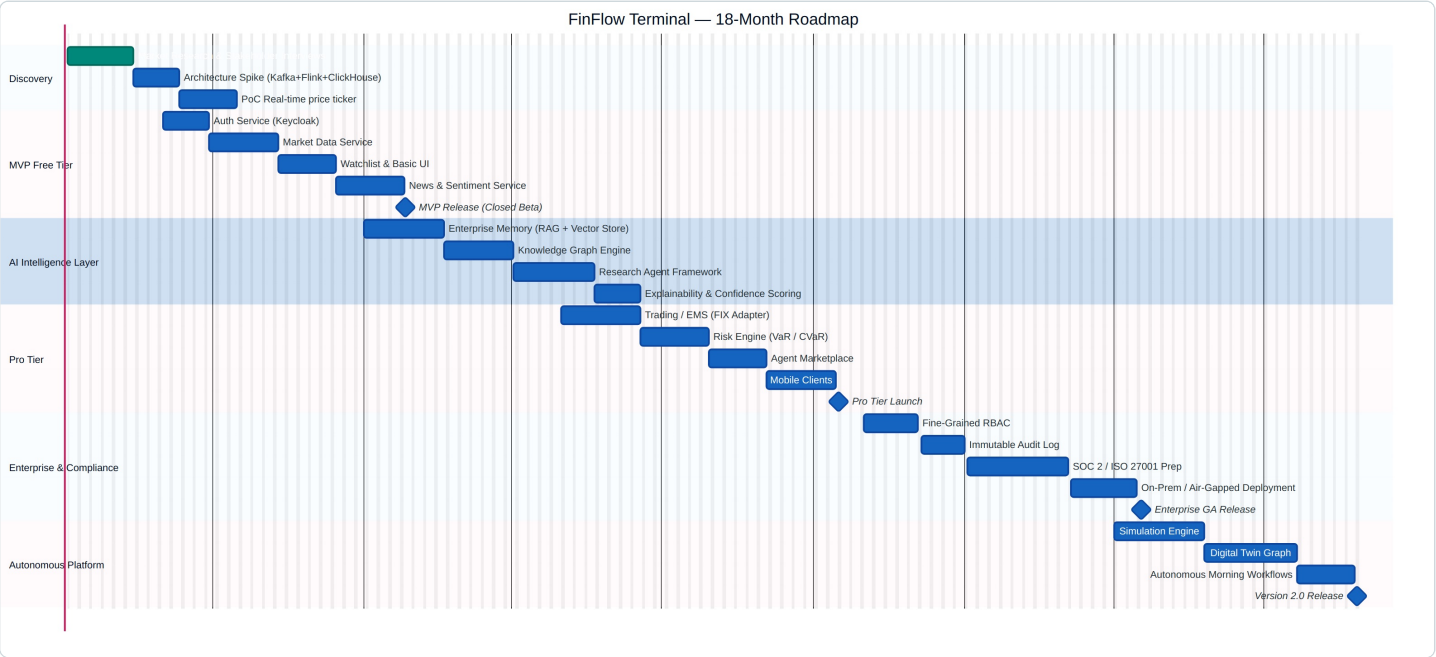


Fig. 10.1 — Revised 18-month roadmap with AI Intelligence Layer (Q1 2027) and Autonomous Platform (Q2 2028) phases inserted.

Milestone	Target	Key Deliverable
M1 — Architecture Spike	2026-07-25	PoC proving end-to-end latency <200 ms for a single asset.
M2 — MVP Free Tier	2026-12-15	Auth, market data, watchlist, news, basic UI.
M3 — AI Intelligence Layer NEW	2027-03-15	Enterprise memory, knowledge graph, first 8 research agents, explainability framework.
M4 — Pro Tier	2027-08-31	Trading/EMS, risk engine, agent marketplace, mobile apps.
M5 — Enterprise GA	2028-02-28	RBAC, immutable audit log, GDPR/CCPA, SOC 2/ISO 27001, on-prem & air-gapped deployment.
M6 — Autonomous Platform NEW	2028-07-31	Simulation engine, digital twin, autonomous morning workflows.
M7 — Version 2.0 Release	2028-09-15	Full feature set, performance SLA met, documentation complete.

11. Assumptions, Dependencies & Constraints

#	Assumption / Dependency	Impact if Violated
A1	Kafka cluster sustains ≥200k msgs/sec peak ingest.	Larger broker fleet or asset-class sharding; higher cost.
A2	ClickHouse provides sub-second analytical queries on billions of rows.	Consider Druid/Pinot; schema changes.
A3	Regulatory sandbox (FCA, SEC) available for trading feature testing.	Delay live trading launch; limit to simulated mode.
A4	Third-party data licensors grant reasonable wholesale rates.	Higher OPEX; tighter BYOD focus.
A5	Team has JVM (Kafka/Flink), Go/Rust, TypeScript, and Python/ML expertise.	Timeline extension; hiring/training needed.
A6 NEW	Underlying LLM providers (or self-hosted models) meet latency and cost targets for multi-agent orchestration.	Fall back to smaller/local models for latency-sensitive agents; degrade gracefully.
A7	Legal counsel confirms Matrix E2EE chat and on-prem LLMs comply with export-control regulations (ITAR).	Restrict encryption strength/model weights or obtain licenses.
A8	Users accept a modern, agent-driven UI over the Bloomberg keyboard/GO-command workflow.	Provide Bloomberg-mode skin and training material as a bridge, not the core UX.

Constraints: CLAUDE.md versioning rule (semver bump + lockfile commit per production change) · ≥90% unit / ≥70% integration coverage before staging promotion · container images ≤500 MB · no cross-provider module imports outside `core/` · every AI answer must carry an evidence chain (Section 15.5) before it can be surfaced to a user.

12. Acceptance Criteria & Verification

Feature	Acceptance Test (summary)	Tool / Method
Login & MFA	Successful auth via password, OIDC, TOTP; failures logged.	Postman + security scan
Market Data Latency	95th-p tick-to-display ≤150 ms under 10k msgs/sec load.	k6 load test + Grafana
Watchlist CRUD	Create/update/delete propagates to clients within 2 s.	Cypress
Trade Order	UI → FIX → broker simulator → filled → persisted → UI reflects status.	QuickFIX/J + DB assertions
Risk VaR Calculation	VaR matches independent calculation within 0.5% tolerance.	Python/numpy/quantlib script
AI Research Agent NEW	For a benchmark set of 100 research questions, agent report cites ≥1 verifiable source per claim and confidence score correlates with human-rated accuracy (Spearman ≥0.6).	Golden-question eval harness + human review panel
Explainability NEW	100% of sampled agent answers include source list, confidence score, and any contradictory evidence found.	Automated schema validation on agent output
Air-Gapped Deployment NEW	Full research-agent workflow completes with zero outbound network calls when deployed in air-gapped mode.	Network policy audit + egress firewall test
Agent Marketplace NEW	Uploaded agent passes signature validation, security scan, and runs in sandbox with no host filesystem access.	Bandit/Semgrep + runtime sandbox test
GDPR Data-Subject Request	Export API returns all personal data; Delete API leaves anonymized audit trail.	API contract test + DB query
Security Pen-Test	No critical/high findings in OWASP ZAP / Burp Suite scan.	CI step + quarterly external pen-test

13. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Data Licensing Cost Overrun	Medium	High	Start with free/open data; negotiate wholesale rates; BYOD shifts cost to customers.

Regulatory Non-Compliance	Low	Very High	Fintech legal counsel from inception; embedded audit-logging and data-subject APIs.
Performance Bottleneck	Medium	High	Per-sprint profiling; kernel-bypass ingest; in-memory hot paths.
Security Breach	Low	Very High	Zero-trust, mTLS, Vault secrets, red/blue-team exercises, bug-bounty.
Adoption Resistance	Medium	Medium	Bloomberg-mode skin, tutorials, trader advisory board.
AI Hallucination / Wrong Investment Advice NEW	Medium	Very High	Mandatory evidence-chain citation, confidence scoring, human-approval gate before any trade-affecting action (Section 15.5, 15.12).
Model / Data Sovereignty Concerns from Large Funds NEW	Medium	High	On-prem, air-gapped, and BYO-model deployment options (Section 15.6).
Category Confusion — "Just Another Bloomberg Clone" NEW	Medium	High	Positioning discipline: never market as "Bloomberg alternative"; lead all messaging with the AI Decision Engine (Section 15.12).
Talent Shortage (Kafka/Flink, ML/agents)	Medium	Medium	Internal training, contractors, managed services (Confluent, AWS MSK/Bedrock).

14. Appendix A — Glossary of Terms

Term	Definition
Asset Class	Group of securities with similar characteristics (equities, fixed-income, FX, commodities, crypto).
FIGI	Financial Instrument Global Identifier.
LEI	Legal Entity Identifier — 20-character code identifying legally distinct parties.
Kafka Topic	A named, durable, ordered log to which records are published.
Flink	Stream-processing framework for stateful computation over bounded/unbounded data.
WORM	Write Once, Read Many — immutable storage model used for audit logs.
RBAC	Role-Based Access Control.
E2EE	End-to-End Encryption.
BYOD	Bring-Your-Own-Data — customers ingest proprietary datasets as first-class sources.
Zero-Trust	Security model assuming no implicit trust; every request authenticated/authorized/encrypted.
RAG NEW	Retrieval-Augmented Generation — LLM responses grounded in retrieved documents/evidence.
Knowledge Graph NEW	Graph database of entities (companies, people, commodities) and typed relationships between them.
Agent Orchestrator NEW	Component that decomposes a natural-language question into sub-tasks and routes them to specialized agents.
Digital Twin NEW	Live, continuously updated simulation model of a company/market ecosystem used for what-if analysis.
Air-Gapped Deployment NEW	Installation with no outbound connection to the public internet; all inference and data stay on customer infrastructure.

15. Strategic Addendum — AI-Native Platform Extensions

Twelve capability areas identified by the architecture review as necessary to move FinFlow from "Bloomberg + AI" to a genuinely new category: the AI Operating System for Global Capital Markets.

15.1 Enterprise Financial Memory

The single largest gap in v1.0.0. Large institutions hold decades of PDFs, research notes, analyst emails, meeting transcripts, Bloomberg exports, and Excel models that are effectively unsearchable today. FinFlow shall provide an **Enterprise Financial Memory**: every ingested document is chunked, embedded, entity-linked into the Knowledge Graph (Section 15.3), and made queryable in natural language alongside live market data.

ID	Requirement
FR-022a	System shall ingest PDFs, DOCX, XLSX, PPTX, email (EML/MSG), and plain text into the memory pipeline.
FR-022b	System shall preserve document-level access control (tenant, team, individual) through to retrieval time — a user shall never retrieve chunks from a document they cannot open.
FR-022c	System shall support semantic + keyword hybrid search with sub-second p95 latency at 10M+ document scale.

15.2 AI Research Agent Network

Rather than a generic notebook, FinFlow ships purpose-built **Research Agents** that compose the investment-research workflow: filings, earnings, patents, macro, news, competitor/supply-chain, valuation (DCF/comps), and quantitative back-testing. A user asks a single question; the Agent Orchestrator decomposes it, dispatches the relevant agents in parallel, and reconciles their findings into one evidence-based report.

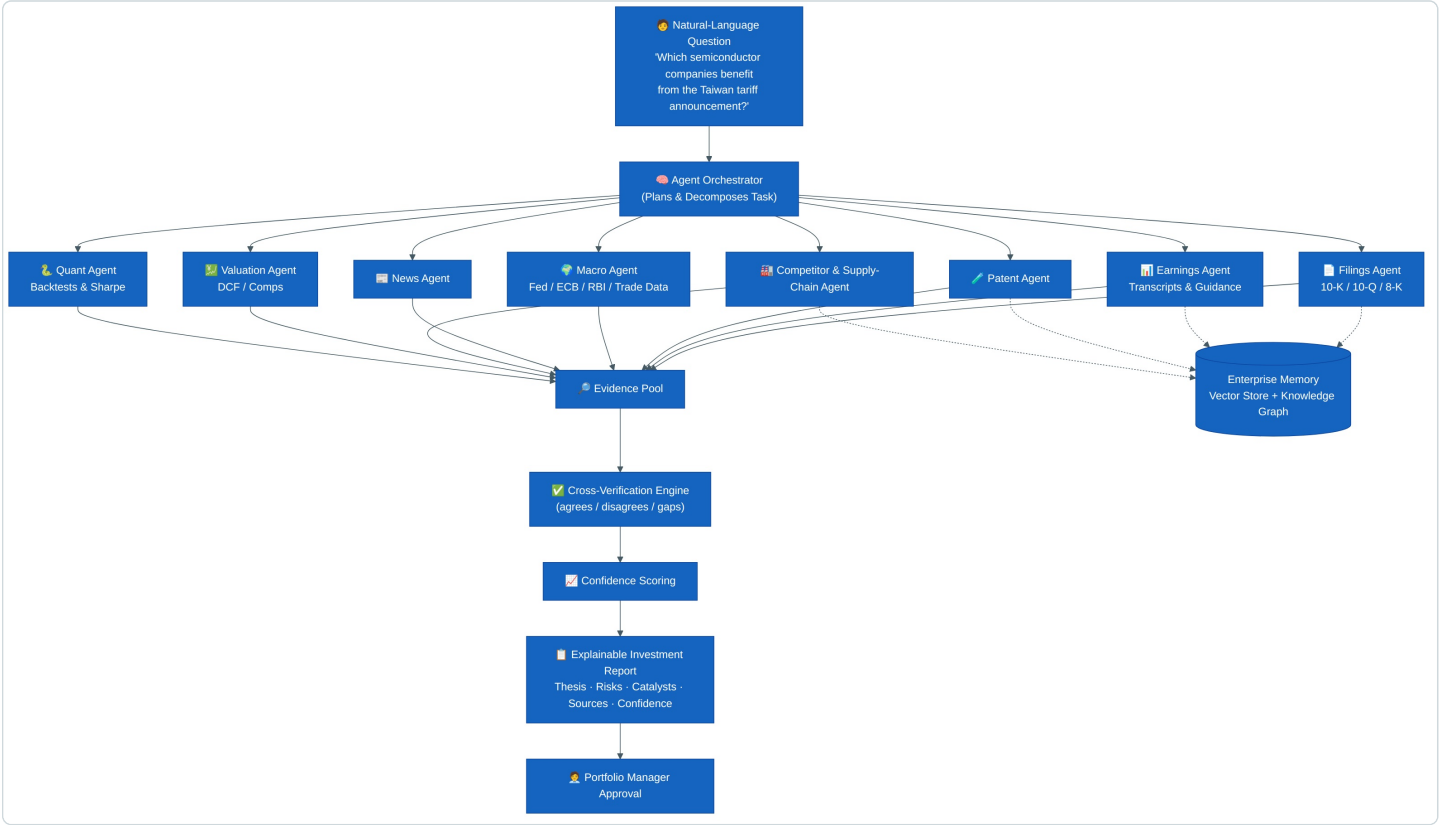


Fig. 15.1 — Agent Orchestrator decomposing a research question across eight specialized agents into a single explainable report.

Produces investment thesis, DCF, risks, catalysts, and competitor comparison.

Macro Agent

Reads central-bank communications, inflation, GDP, employment, and commodity data to generate sector outlooks.

M&A Agent

Cross-references filings, market share, and patents to surface acquisition candidates and rationale.

Quant Agent

Writes and executes backtests, optimizes strategy parameters, and explains resulting Sharpe/Sortino ratios.

15.3 Financial Knowledge Graph

Bloomberg stores data points; FinFlow stores *relationships*. The Knowledge Graph links companies to suppliers, patents, executives, litigation, countries, commodities, shipping routes, and customers, enabling questions like "which suppliers depend on NVIDIA?" to be answered by graph traversal rather than manual research.

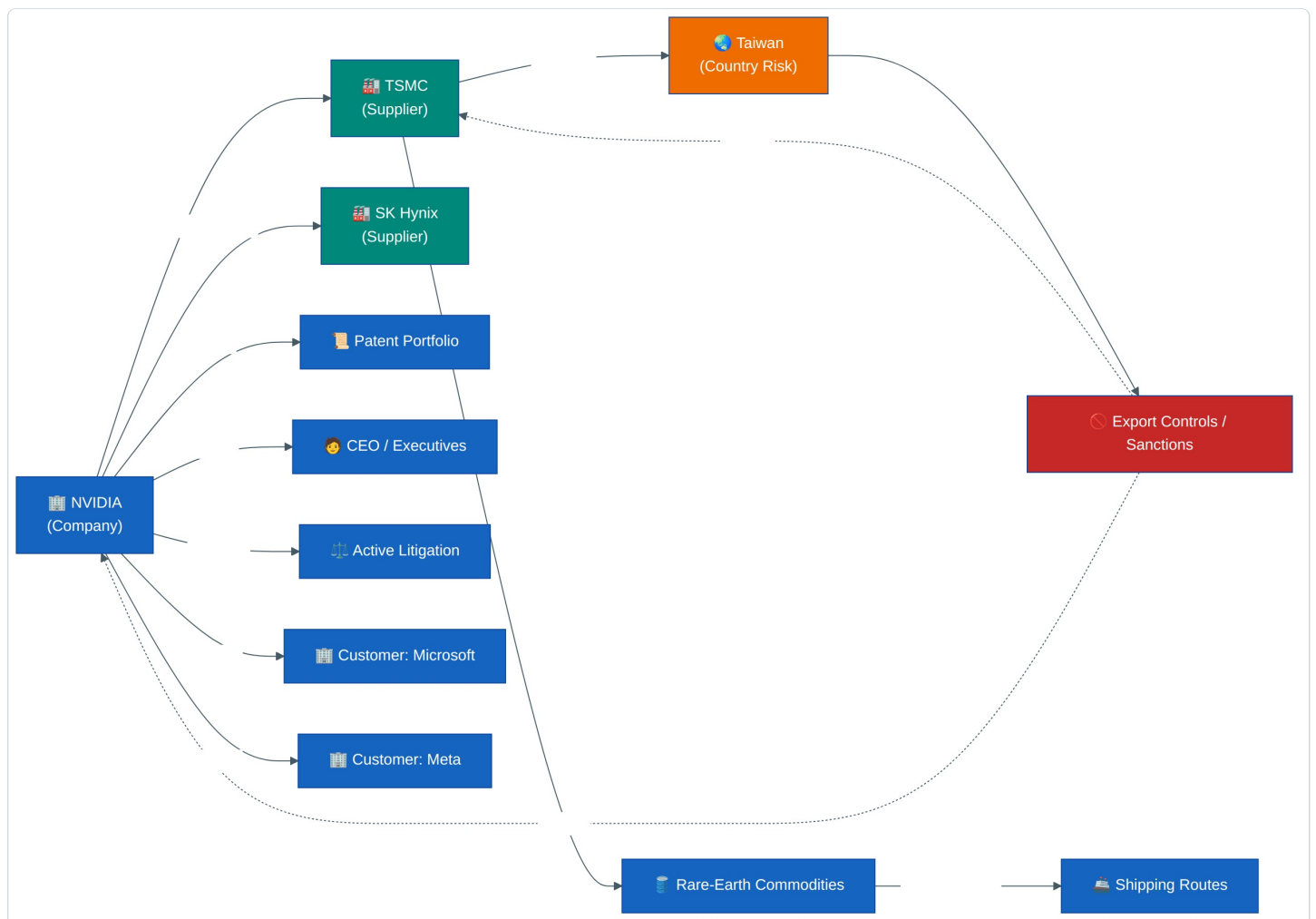


Fig. 15.2 — Sample knowledge-graph slice around a single company, showing supplier, geopolitical, and sanctions relationships.

15.4 Scenario Simulation Engine

Given a shock scenario ("What happens if oil reaches \$120?"), the Simulation Engine propagates the shock across currencies, equities, inflation, bank balance sheets, transportation costs, GDP, and individual company earnings, using the Knowledge Graph to trace second- and third-order effects through supply chains.

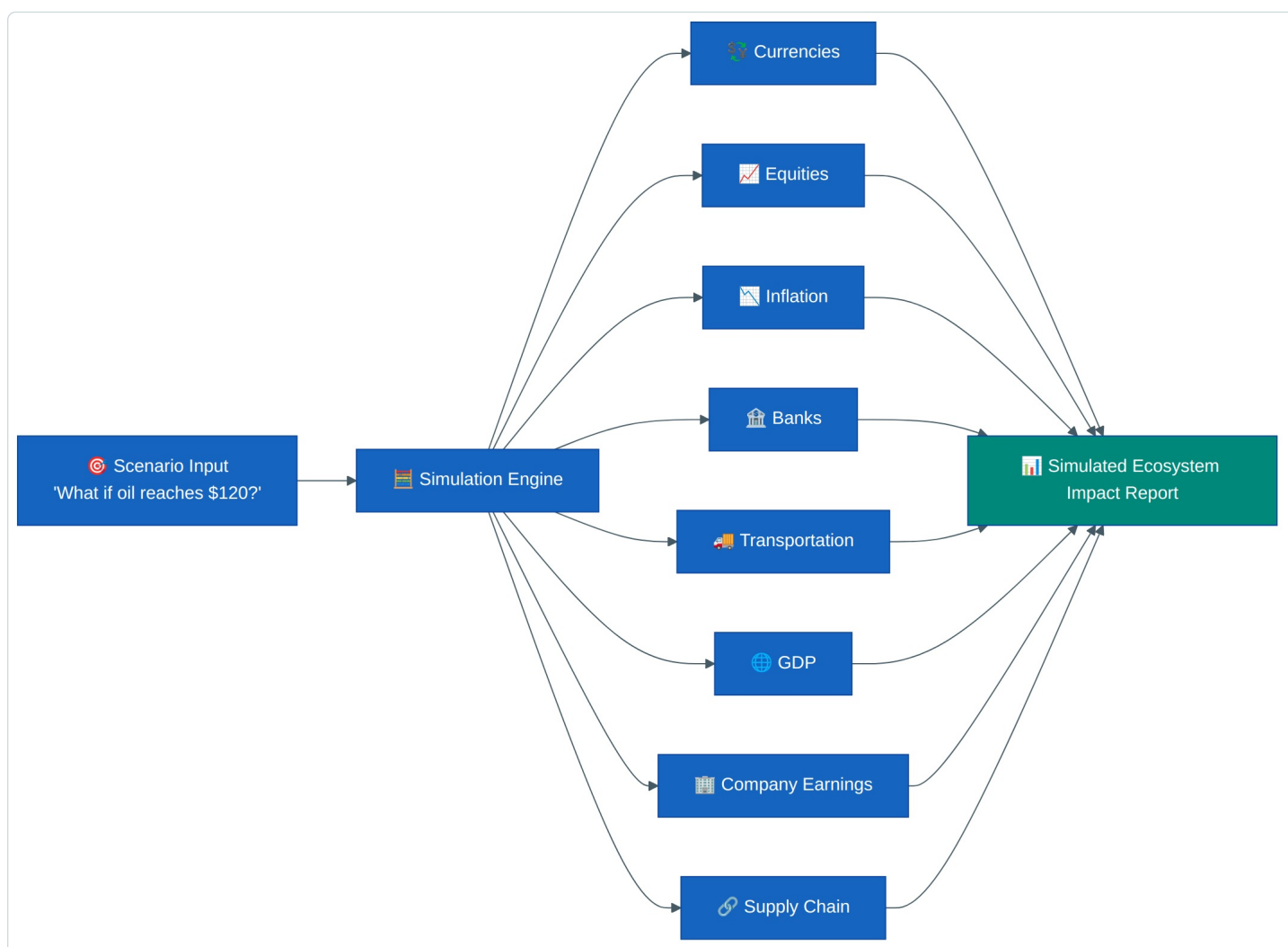


Fig. 15.3 — Scenario input propagating through the simulation engine into a cross-asset impact report.

15.5 Explainable AI Framework

Enterprise customers will not trust black-box output. Every AI-generated answer — from a single price question to a full investment thesis — shall carry, as machine-readable metadata: source documents, a confidence score, the chain of evidence used, any contradictory evidence found, explicitly flagged missing information, and data freshness timestamps. This is enforced at the API contract level (FR-025), not left to individual agents' discretion.

15.6 Sovereign & On-Prem AI Deployment

Large hedge funds and banks will not send proprietary data to a third-party model they do not control. FinFlow shall support on-prem inference, air-gapped installation with zero outbound network calls, and bring-your-own-model configurations (Llama, Qwen, DeepSeek, Claude via API, Gemini, OpenAI), selectable per tenant.

15.7 Agent Marketplace

The original Plugin Marketplace is reframed as an **Agent Marketplace** — a "GitHub for Finance" where third parties publish signed, sandboxed Valuation, Energy, Healthcare, Fraud, ETF, Macro, and Commodity agents, versioned and security-scanned before publish, with usage-based revenue share.

15.8 Autonomous Workflows

The platform shall support fully unattended scheduled workflows: overnight ingestion of news, filings, and earnings; model and DCF recalculation; anomaly detection against the prior day's positions; and an automatically generated, explainable morning brief delivered to the portfolio manager before market open — with no human touching any step until final review.

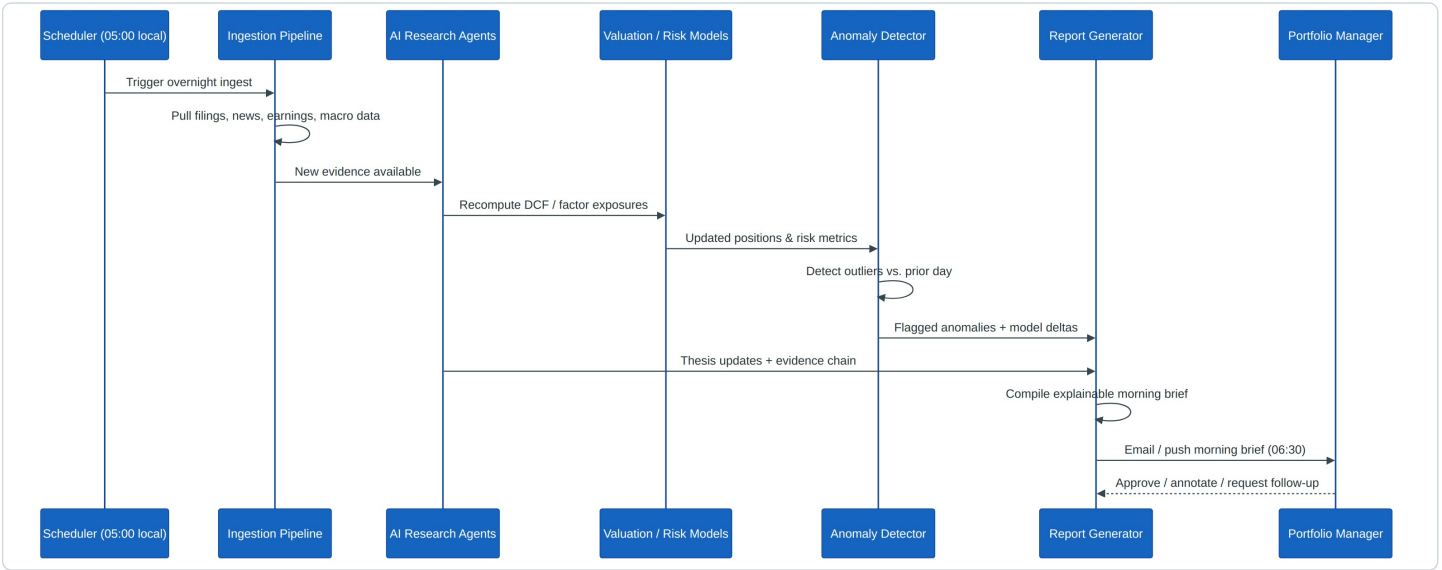


Fig. 15.4 — Autonomous 05:00 morning-brief workflow, from overnight ingestion to portfolio-manager approval.

15.9 Voice & Conversational Interface

In addition to the Bloomberg-style command palette (FR-017, now a supporting feature), FinFlow shall accept natural spoken queries — e.g., "show me companies that will benefit from India's semiconductor policy" — and respond conversationally, routing through the same Agent Orchestrator as typed queries.

15.10 Multi-Modal Financial Ingestion

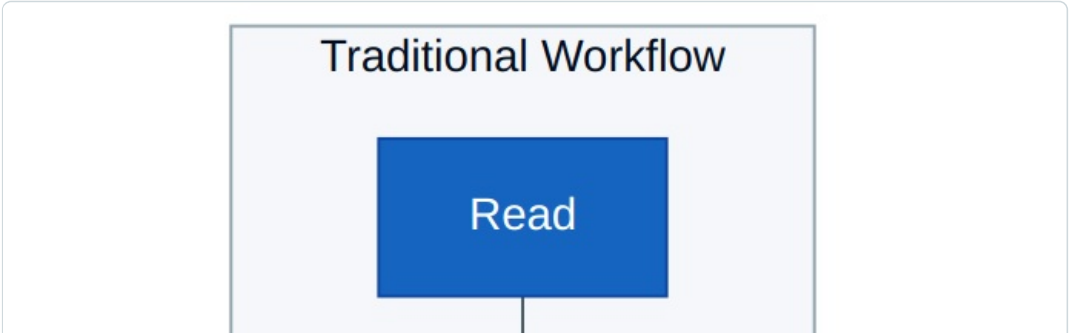
Users shall be able to upload images, PDFs, spreadsheets, slide decks, or earnings-call recordings and have the platform reason across all of them simultaneously — e.g., cross-referencing a chart image against a filing's stated figures.

15.11 Financial Digital Twin

Extending the Simulation Engine, FinFlow shall maintain a continuously updated live graph of a company's products, factories, suppliers, ports, employees, customers, shipping routes, patents, sanctions exposure, weather risk, and commodity dependencies, so that "what happens if Taiwan stops exports?" updates the entire modeled ecosystem rather than a single data point.

15.12 Enterprise Decision Engine

This is the central repositioning of the product. The traditional research workflow — read, research, think, model, present, decide — is compressed into a single orchestrated pipeline: a question triggers AI decomposition, 50+ agents investigate in parallel, findings are cross-verified, a valuation and simulation are built, an explainable report is produced, and a human makes the final call.



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graph TD; Start(( )) --> Research[Research]; Research --> Think[Think]; Think --> Model[Model]; Model --> Present[Present]; Present --> Decide[Decide];
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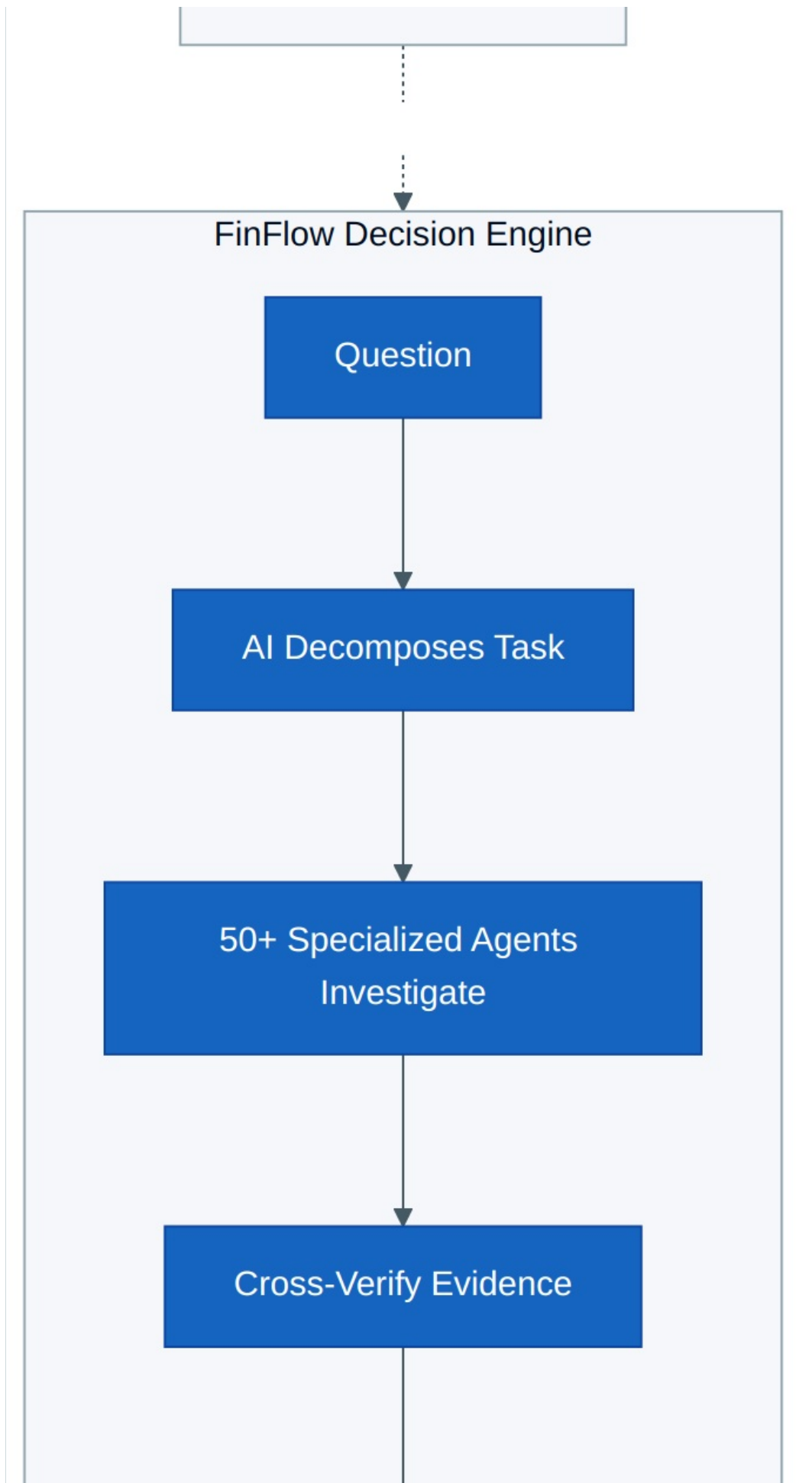
Research

Think

Model

Present

Decide



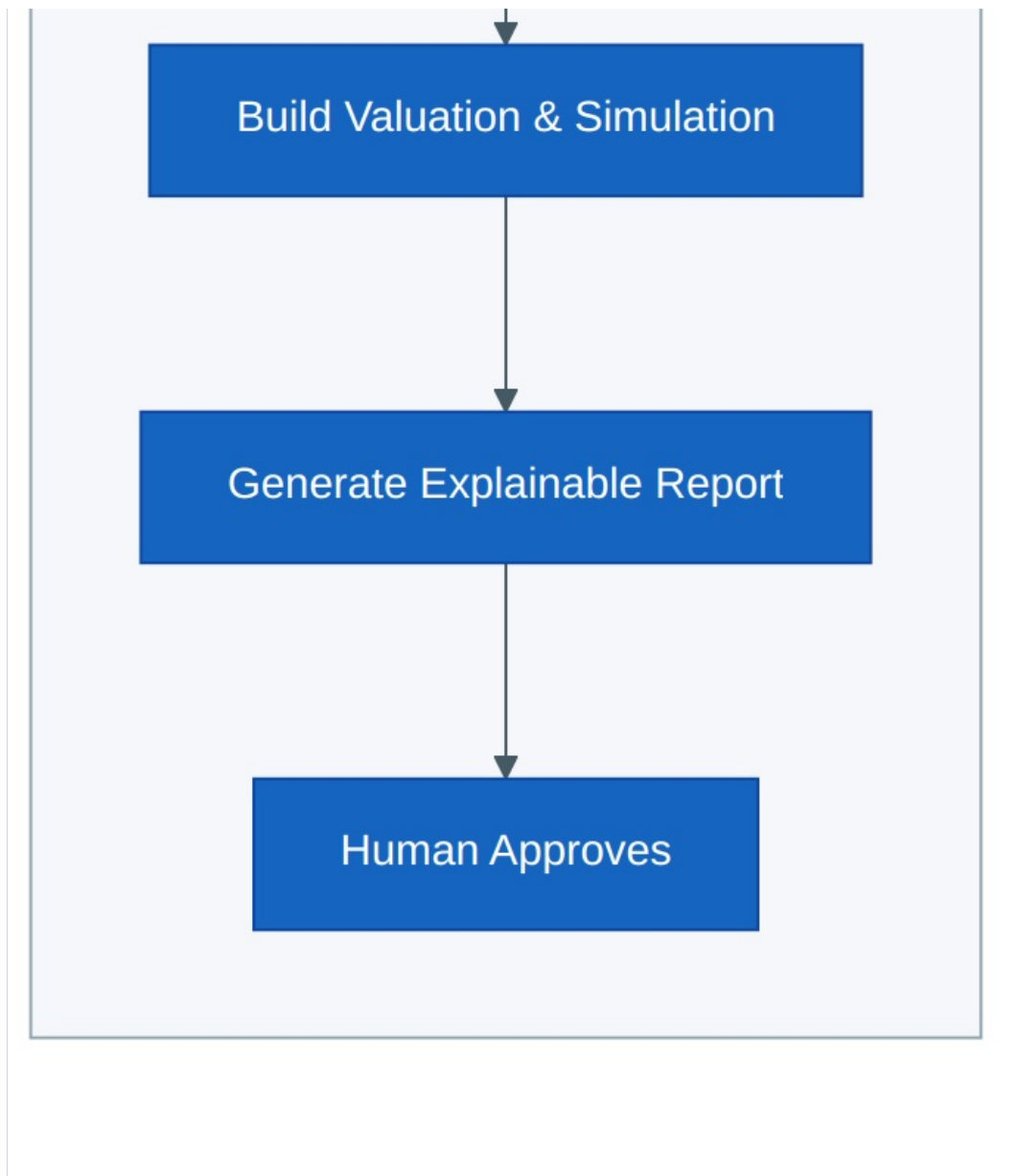


Fig. 15.5 — Traditional research workflow (top) reimagined as the FinFlow Enterprise Decision Engine (bottom).

Positioning note: the platform is never marketed as a "Bloomberg alternative." Approved positioning statements are: *"The AI Operating System for Global Capital Markets"* and *"The world's first autonomous financial intelligence platform."*

15.13 Revised Functional Requirements Table

See FR-021 through FR-029 in Section 6.1 for the formal, testable requirements corresponding to Sections 15.1–15.10.

15.14 Revised Roadmap Rationale

Phase	Rationale for Placement
AI Intelligence Layer (Q1 2027, after MVP)	Enterprise Memory and the first agent set need a stable market-data and reference-data foundation (M2) before they have anything reliable to reason over.
Agent Marketplace (Pro Tier, alongside Trading/EMS)	Third-party agent publishing requires the sandboxing and billing infrastructure already scheduled for the Plugin Marketplace slot — reused, not duplicated.
Sovereign/On-Prem AI (Enterprise GA)	Bundled with existing on-prem Helm chart and SOC 2/ISO 27001 workstreams to avoid a second infrastructure hardening pass.

Simulation Engine & Digital Twin (Autonomous Platform, final phase)	Depends on a mature Knowledge Graph and a large enough Enterprise Memory corpus to model second-order supply-chain effects credibly.
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16. Appendix B — Architecture Review Summary

Summary of the internal review that produced this revision (full notes retained by the Architecture Team).

Dimension	Score	Note
Architecture quality	9.5 / 10	Microservice topology, event backbone, and data model judged sound and scalable.
Enterprise readiness	9 / 10	Compliance, audit, RBAC, and HA design already enterprise-grade.
Technical depth	10 / 10	No material gaps found in the core platform engineering.
Business strategy (original)	5.5 / 10	Positioned as a Bloomberg alternative — a path multiple prior entrants (Refinitiv, FactSet, Koyfin, Symphony, AlphaSense, Sentio) have already taken without displacing the incumbent.

Core finding: hedge funds and asset managers do not need another chart or another feed — they are already saturated with data (50+ earnings reports, 100+ filings, dozens of macro and research reports, and thousands of news items per day). What they lack is a system that turns that volume into a specific, sourced, confidence-scored answer. That gap — not feature parity with Bloomberg — is the addressable opportunity, and it is what Section 15 is designed to close.

Approximately 70% of the original v1.0.0 architecture remains valid under the revised strategy; the change is primarily one of product framing and the addition of the AI Intelligence Layer, not a rebuild of the core platform.